

Web Server Availability Incident & SYN Flood Analysis

1. Project overview

Conducted a simulated cybersecurity incident investigation involving a web server availability disruption caused by abnormal TCP SYN traffic patterns.

The project focused on:

- denial-of-service attack analysis,
- Incident response procedures,
- Availability risk assessment,
- Network traffic investigation,
- Mitigation planning.

The investigation involved identifying malicious traffic behavior affecting a public-facing company website and documenting the operational impact on users and business services.

2. Incident scenario

A company website became unavailable and users experienced repeated connection timeout errors while attempting to access organizational services.

Network traffic analysis identified an unusually high volume of TCP SYN requests originating from suspicious IP activity, overwhelming the web server and preventing legitimate user connections.

3. Investigation activities

- investigated web server connectivity failures,
- Reviewed packet-sniffing and network traffic data,
- Identified abnormal TCP SYN request volume,
- Analyzed TCP three-handshake behavior,
- Determined indicators consistent with SYN flooding
- Evaluated impact on service availability and users
- Documented incident findings and attack behavior
- Assessed short-term mitigation strategies.

4. Key findings

The investigation identified:

- excessive inbound TCP SYN traffic,
- Web server resource exhaustion,
- Inability to establish legitimate client connections,
- Degraded website availability,
- Connection timeout errors affecting users.

The incident demonstrated how SYN flood attacks can disrupt business operations by exhausting server resources required to complete legitimate TCP sessions.

5. Mitigation & Response actions

Response activities included:

- temporarily isolating the affected server,
- Implementing firewall-based IP blocking,
- Escalating the issue for further investigation,
- Assessing limitations of static IP blocking due to spoofing risks,
- Recommending additional network protection measures.

6. Governance & Risk Perspective

The project emphasized:

- availability risk management,
- Incident escalation procedures,
- Service continuity concerns,
- Operational impact assessment,
- Security monitoring effectiveness,
- Network attack identification,
- Response coordination between security and IT teams.

7. Recommendations

Recommended next steps included :

- implementing SYN-flood protection mechanism,
- Improving traffic monitoring and alerting,
- Reviewing firewall and rate-limiting configurations,
- Deploying intrusion detection/prevention systems,
- Strengthening denial-of-service response procedures.

8. Skill demonstrated

- Incident analysis,
- DoS/SYN-flood Identification,
- Network Traffic Analysis,
- TCP/IP Fundamentals,
- availability risk assessment,
- Incident documentation,
- Mitigation planning,
- Firewall response actions,
- Operational risk awareness,
- Security monitoring,
- Cybersecurity reporting.